

## Executive Summary

This report documents a complete end-to-end AML (Anti-Money Laundering) analytics pipeline built on the IBM HI-Small synthetic transaction dataset. The project spans raw data ingestion through to a four-page compliance dashboard in Power BI, with all analytical logic written in SQL and all documentation maintained for reproducibility.

The objective was to replicate the analytical workflow of a GRC or Compliance Analyst: monitor transaction flows, detect laundering typologies, score entity risk, and enable investigation-ready drill-through views aligned with SAR preparation requirements.

### Dataset at a Glance

|                                        |                                      |                                 |                                       |
|----------------------------------------|--------------------------------------|---------------------------------|---------------------------------------|
| <b>5M</b><br>Total Transactions        | <b>5K</b><br>Suspicious Transactions | <b>0.10%</b><br>Laundering Rate | <b>\$22.90T</b><br>Total Volume (USD) |
| <b>\$187.07bn</b><br>Suspicious Volume | <b>212</b><br>High Risk Accounts     | <b>603</b><br>Fan-Out Accounts  | <b>20</b><br>Days of Data             |

### Key Findings

- ACH is the highest-risk payment format at 0.75% laundering rate — 7.5x the dataset average of 0.10%.
- Saudi Riyal same-currency transactions have the highest laundering rate of any currency pair at 0.42%.
- Sunday activity shows the highest laundering concentration, peaking between hours 9 and 12.
- 603 accounts exhibit fan-out patterns with 5 or more unique counterparties, a key layering indicator.
- Bank 4 originates the highest suspicious outbound volume at \$84.86bn despite a below-average laundering rate of 0.08% — indicating scale exposure.
- Bank 142183 has the highest laundering rate among all banks at 0.91%.

## Project Methodology

The project followed a four-stage pipeline: Extract, Transform, Load, Visualise (ETLV). Each stage is documented and reproducible.

| Stage        | Tool             | Action                                                                                                                   | Output                                   |
|--------------|------------------|--------------------------------------------------------------------------------------------------------------------------|------------------------------------------|
| 1. Extract   | DuckDB CLI       | Read HI-Small_Trans.csv into staging table using read_csv_auto()                                                         | transactions_stage table                 |
| 2. Transform | DuckDB SQL       | Cast types (DECIMAL for amounts, TIMESTAMP, BOOLEAN), rename duplicate Account columns, assign row IDs                   | Final transactions table (5M typed rows) |
| 3. Cache     | DuckDB SQL       | Pre-aggregate into 6 cache tables covering daily summaries, entity risk, heatmap, bank risk, and suspicious transactions | 6 cache tables (11 to 5K rows each)      |
| 4. Export    | DuckDB COPY      | Export each cache table to CSV using COPY ... TO with HEADER                                                             | 6 CSV files for Power BI                 |
| 5. Visualise | Power BI Desktop | Import CSVs in Import mode, build data model, create DAX measures, design 4-page dashboard                               | AML_Dashboard.pbix                       |
| 6. Deliver   | GitHub Desktop   | Commit all SQL, documentation, screenshots, and dashboard file to public repository                                      | Public GitHub portfolio repository       |

### Why DuckDB Over MySQL Workbench

The original pipeline was designed around MySQL 8.0 and the Workbench Table Data Import Wizard. At 5 million rows, the Import Wizard took 3–5 hours per attempt and caused repeated connection loss errors ('Error Code: 2013: Lost connection to MySQL server during query'). Even with correct session settings (max\_allowed\_packet, wait\_timeout), the GUI overhead made large-scale analytical queries unstable.

| Dimension                  | MySQL Workbench                     | DuckDB (Final)                |
|----------------------------|-------------------------------------|-------------------------------|
| CSV Import (5M rows)       | 3–5 hours                           | 30–60 seconds                 |
| Analytical query stability | Frequent timeouts                   | No timeouts — embedded engine |
| Setup required             | Server config, ODBC, session tuning | Single CLI command            |
| Power BI connection        | Live queries — slow on 5M rows      | CSV import — instant load     |
| Crashes on Mac             | Yes — repeated                      | Never                         |

### Cache Table Architecture

Rather than querying the raw 5-million-row transactions table from Power BI (which causes repeated full-table scans), all analytics are pre-computed in DuckDB and stored as small summary tables. Power BI imports these summary tables as static CSVs.

| Cache Table                   | Rows   | Content                                            | Powers (Dashboard)         |
|-------------------------------|--------|----------------------------------------------------|----------------------------|
| daily_summary_cache           | ~70    | Daily aggregates by payment format                 | Page 1 trend charts        |
| payment_format_cache          | 8      | Risk rate per payment channel                      | Pages 1 & 2 format visuals |
| bank_risk_cache               | ~15    | Volume and laundering rate per bank                | Pages 1 & 3 bank tables    |
| account_risk_cache            | 500    | Top 500 highest-risk accounts with composite score | Pages 2 & 3 entity tables  |
| hourly_heatmap_cache          | ~70    | Laundering rate by day-of-week and hour            | Page 2 heatmap             |
| suspicious_transactions_cache | ~5,000 | All confirmed suspicious transactions              | Page 4 drill-through       |

# Typology Analysis Findings

Four core laundering typologies were implemented as detection logic in DuckDB SQL. Each is aligned to FATF guidance and produces findings visible in the Power BI dashboard.

## 1. ACH Channel Dominance (Placement & Layering)

ACH transactions account for 86.59% of all suspicious transactions by count and carry a laundering rate of 0.75% — 7.5 times the dataset average. This concentration in a single payment channel is a strong indicator of coordinated or automated placement activity.

| Payment Format | Laundering Rate | Suspicious Txns | Risk Assessment                       |
|----------------|-----------------|-----------------|---------------------------------------|
| ACH            | 0.75%           | 86.59% of total | Critical — primary laundering channel |
| Bitcoin        | 0.04%           | 1.08% of total  | Low                                   |
| Cheque         | 0.02%           | 6.26% of total  | Low                                   |
| Credit Card    | 0.02%           | Minimal         | Low                                   |
| Cash           | 0.02%           | Minimal         | Low                                   |

## 2. Currency Pair Analysis (Layering via Conversion)

Same-currency transactions dominate the suspicious activity, with Saudi Riyal-to-Riyal pairs showing the highest laundering rate at 0.42%. This is counterintuitive and may indicate that currency conversion is not being used as a layering mechanism in this dataset. Instead, high-frequency same-currency transfers are the primary pattern.

| Currency Pair             | Laundering Rate | Implication                                |
|---------------------------|-----------------|--------------------------------------------|
| Saudi Riyal → Saudi Riyal | 0.42%           | Highest risk pair — same-currency layering |
| Euro → Euro               | 0.12%           | Above average — monitor EUR flows          |
| US Dollar → US Dollar     | 0.10%           | Average rate — baseline risk               |
| Yen → Yen                 | 0.10%           | Average rate                               |
| Remaining pairs           | 0.09%           | Below average — lower risk                 |

## 3. Temporal Patterns (Heatmap Analysis)

The day-and-hour heatmap reveals a clear temporal pattern: laundering activity is concentrated on weekends, particularly Sundays, with rates peaking between hours 9 and 12. Weekday activity (especially Thursday and Friday) shows the lowest rates. This pattern is consistent with reduced oversight and monitoring during weekend periods.

- Sunday 9:00–12:00 AM: rates reaching 0.53–0.56%. 5x the weekday average
- Saturday rates are elevated throughout the afternoon and evening
- Thursday has the lowest sustained laundering rate of any day
- Peak hours (9–13) on weekdays still show elevated rates on Mondays and Tuesdays

## 4. Fan-Out / Pass-Through Accounts (Layering)

603 accounts were identified with fan-out patterns. This is defined as accounts involved in suspicious transactions with 5 or more unique counterparties. This is a classic layering indicator, where a single account receives funds and distributes them to many receivers to obscure the trail.

- Fan-out accounts: 603
- Definition: suspicious\_txns > 0 AND unique\_counterparties >= 5
- These accounts represent priority targets for further investigation
- Cross-referencing these accounts with the entity investigation page reveals their full risk profile

## Entity Risk Analysis

### Account Risk Scoring Methodology

Each account in the dataset is scored on a 0–100 composite risk scale using four weighted components:

| Component                | Weight           | Rationale                                                |
|--------------------------|------------------|----------------------------------------------------------|
| Suspicion rate %         | Primary driver   | Direct ratio of suspicious to total transactions         |
| Payment format diversity | +5 per format    | Using multiple channels indicates deliberate obfuscation |
| Transaction volume       | +10 if > 50 txns | High-frequency accounts present greater exposure         |
| Cross-currency activity  | +2 per instance  | Currency conversion adds layering complexity             |

### Risk Tier Definitions

| Tier     | Score Range | Action                                            |
|----------|-------------|---------------------------------------------------|
| Critical | 90–100      | Immediate escalation — SAR filing consideration   |
| High     | 70–89       | Priority review — enhanced due diligence required |
| Medium   | 40–69       | Monitoring — periodic review recommended          |
| Low      | 0–39        | Standard monitoring — no immediate action         |

### Top Accounts by Suspicious Transactions

| Account ID | Bank   | Susp. Txns | Susp. Rate | Risk Score | Risk Tier |
|------------|--------|------------|------------|------------|-----------|
| 100428660  | 70     | 243        | 0.14%      | 25.14      | Low       |
| 1004286A8  | 70     | 158        | 0.15%      | 25.15      | Low       |
| 812D22980  | 150240 | 25         | 92.59%     | 100.00     | Critical  |
| 8021353D0  | 23691  | 21         | 95.45%     | 100.00     | Critical  |
| 80266F880  | 15231  | 29         | 29.59%     | 100.00     | Critical  |
| 811C597B0  | 119    | 21         | 56.76%     | 84.16      | High      |

*Note: Account 100428660 has the highest raw suspicious count (243) but a low risk score (25.14) because its suspicion rate is only 0.14% across 168,672 total transactions. Accounts like 812D22980 score 100 with only 25 suspicious transactions because 92.59% of their activity is flagged which is a much stronger risk signal per transaction.*

### Bank-Level Risk Summary

| Bank   | Outbound Txns | Susp. Txns | Laundering Rate | Outbound Volume | Susp. Volume |
|--------|---------------|------------|-----------------|-----------------|--------------|
| 4      | 22,116        | 17         | 0.08%           | \$254.50bn      | \$84.86bn    |
| 14381  | 1,348         | 3          | 0.22%           | \$84.47bn       | \$66.41bn    |
| 142183 | 110           | 1          | 0.91%           | \$2.50bn        | \$2.49bn     |
| 70     | 449,859       | 633        | 0.14%           | \$937.85bn      | \$395.49M    |
| 23842  | 5,910         | 18         | 0.30%           | \$450.80bn      | \$288.82M    |

## Dashboard Overview

The Power BI dashboard is structured as four pages, each designed for a specific compliance audience and use case.

| Page | Title                         | Primary Audience            | Key Visuals                                               |
|------|-------------------------------|-----------------------------|-----------------------------------------------------------|
| 1    | Executive Summary             | Compliance Officer / CISO   | KPI cards, daily trend, bank volume, payment format donut |
| 2    | Typology & Pattern Analysis   | MLRO / AML Analyst          | Heatmap, currency pairs, risk scatter, format risk bars   |
| 3    | Entity Investigation          | Transaction Monitoring Team | Account risk leaderboard, bank risk summary               |
| 4    | Suspicious Transaction Detail | Case Manager / SAR Preparer | Transaction log, account summary cards (drill-through)    |

## Technical Configuration

- Data source: 6 pre-aggregated CSV files exported from DuckDB
- Import mode: Static import (not DirectQuery) — optimal for historical datasets
- Data model: Star schema with DateTable related to daily\_summary\_cache and suspicious\_transactions\_cache
- DAX measures: 15+ measures including Laundering Rate %, High Risk Accounts, Pass-Through Accounts, Cross Currency Rate %
- Interactivity: Synced slicers (date, bank, payment format) across pages 1–3
- Drill-through: Right-click any account on Page 3 to open filtered Page 4
- Conditional formatting: Traffic-light colour coding on risk scores (red/orange/green)
- Theme: Custom AML Compliance JSON theme (dark green palette, compliance-appropriate)

## DAX Risk Measures — Key Definitions

| Measure               | Formula Logic                                                        |
|-----------------------|----------------------------------------------------------------------|
| Laundering Rate %     | DIVIDE([Suspicious Transactions], [Total Transactions], 0) × 100     |
| High Risk Accounts    | COUNTROWS of account_risk_cache WHERE risk_score >= 75               |
| Pass-Through Accounts | COUNTROWS WHERE suspicious_txns > 0 AND unique_counterparties >= 5   |
| Suspicious Volume %   | DIVIDE([Suspicious Volume USD], [Total Volume USD], 0) × 100         |
| Cross Currency Rate % | DIVIDE([Cross Currency Transactions], [Total Transactions], 0) × 100 |

# Technical Challenges & Resolutions

Several significant technical challenges were encountered and resolved during this project. Each is documented here as a reference for future projects.

## Challenge 1: MySQL Workbench Import Failure

| Item       | Detail                                                                                                                                                                    |
|------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Problem    | Table Data Import Wizard took 3–5 hours per attempt and repeatedly crashed with 'Error Code: 2013: Lost connection to MySQL server during query'                          |
| Root cause | MySQL Workbench parses CSV row-by-row through the GUI layer with no batch optimisation. Default packet sizes and timeouts are insufficient for 5M rows even after tuning. |
| Resolution | Migrated entirely to DuckDB. Import time dropped from 3–5 hours to 30–60 seconds using read_csv_auto(). No configuration required.                                        |
| Lesson     | For analytical workloads on large flat files, embedded OLAP databases (DuckDB) are significantly more appropriate than traditional RDBMS with GUI tools.                  |

## Challenge 2: MySQL PATH Not Found in Terminal

| Item       | Detail                                                                                                                           |
|------------|----------------------------------------------------------------------------------------------------------------------------------|
| Problem    | Terminal returned 'zsh: command not found: mysql' when attempting LOAD DATA INFILE as a faster import alternative                |
| Root cause | MySQL was installed via DMG (not Homebrew), so the binary was not added to the system PATH automatically                         |
| Resolution | Located binary at /usr/local/mysql-8.0.45-macos15-arm64/bin/mysql and used full path. Ultimately superseded by DuckDB migration. |

## Challenge 3: Analytical Query Timeouts (MySQL)

| Item       | Detail                                                                                                                                                                                                                                          |
|------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Problem    | Complex analytical queries (account risk scoring, pass-through detection) timed out or returned 'Lost connection' errors in MySQL Workbench                                                                                                     |
| Root cause | Full-table scans on 5M rows with window functions and multiple self-joins exceeded available RAM and session timeout limits on the Mac hardware                                                                                                 |
| Resolution | Rewrote all queries with pre-filtering to is_laundering=1 first (reducing working set to ~5K rows), used temporary tables for intermediate aggregation, and added LIMIT clauses. This reduced execution time from 10+ minutes to 20–30 seconds. |
| Lesson     | Always filter before aggregation. Scanning 5K rows instead of 5M rows is a 1000x reduction in work — achievable with a single WHERE clause on an indexed column.                                                                                |

## Challenge 4: Duplicate Column Headers in CSV

| Item       | Detail                                                                                                                                                                         |
|------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Problem    | The HI-Small CSV has two columns named 'Account' (one for origin, one for destination). This caused import failures in MySQL Workbench and type-detection confusion in DuckDB. |
| Resolution | DuckDB auto-renames the second instance to 'Account_1'. The final typed table maps these explicitly to from_account and to_account with clean column names.                    |

## Regulatory & Compliance Context

This project was designed with the following regulatory frameworks in mind. The dashboard pages and detection logic map directly to compliance obligations.

| Framework        | Relevant Requirement           | Dashboard Mapping                                                             |
|------------------|--------------------------------|-------------------------------------------------------------------------------|
| FATF Rec. 20     | Report suspicious transactions | Page 4 (Suspicious Transaction Detail) is designed for SAR case building      |
| FATF Rec. 10     | Customer Due Diligence         | Page 3 entity risk scoring surfaces accounts requiring enhanced due diligence |
| BSA / FinCEN CTR | \$10,000 reporting threshold   | Structuring detection queries identify sub-threshold clustering patterns      |
| EU AMLD6         | Expanded predicate offences    | Cross-currency and cross-border transaction analysis on Page 2                |
| Basel AML Index  | Institutional risk indicators  | Bank-level risk table on Page 3 mirrors institutional risk scoring            |

### SAR Preparation Alignment

The Suspicious Transaction Detail page (Page 4) is specifically designed to support Suspicious Activity Report preparation. The information presented maps to standard SAR fields:

- Transaction timestamps — SAR field: date of suspicious activity
- From/To account and bank identifiers — SAR field: subject information
- Amount paid and received — SAR field: amount involved
- Payment currency — SAR field: transaction type
- Is cross-currency flag — SAR field: international transaction indicator
- Account summary KPIs — SAR field: total activity during review period

## Skills Demonstrated

| Skill Area       | Specifics                                                                                                                                                                                 |
|------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Data Engineering | 5M row CSV import, type casting with DECIMAL(18,2) for monetary data, staging table pattern, cache table architecture, CSV export workflow                                                |
| SQL Analytics    | Typology detection queries, composite risk scoring, window functions (row_number, OVER), aggregation with GROUP BY, HAVING, pre-filtering optimisation for large datasets                 |
| DuckDB           | read_csv_auto(), strptime() for timestamp parsing, COPY ... TO for CSV export, persistent database files, embedded OLAP workflows                                                         |
| Power BI / DAX   | Import mode vs DirectQuery, data model relationships, DateTable creation, DIVIDE/CALCULATE/FILTER/COUNTROWS measures, conditional formatting, drill-through configuration, slicer syncing |
| Financial Crime  | AML typologies (structuring, layering, fan-out), regulatory context (FATF, BSA, AMLD6), SAR-aligned investigation views, entity risk scoring, CTR threshold analysis                      |
| Documentation    | Six structured Markdown guides with step-by-step reproducibility, compliance context embedded throughout, checklists, decision trees, and performance comparison tables                   |
| Problem Solving  | Diagnosed and resolved MySQL Workbench import failures, PATH issues on macOS, query timeout errors, and duplicate column header problems. adapting the pipeline at each stage             |

## Portfolio Value

This project demonstrates end-to-end capability across the full analytics stack relevant to GRC and financial crime roles: from understanding raw compliance data through to a professional dashboard a Compliance Officer could use in a real monitoring workflow. The combination of SQL pattern detection, entity risk scoring, and SAR-aligned investigation views directly mirrors what transaction monitoring teams produce in practice.

## Tools Used

- DuckDB 1.0+ — Embedded OLAP database for data import and analytics
- Power BI Desktop — Dashboard and visualisation
- GitHub Desktop — Version control and portfolio delivery
- macOS (Apple Silicon) — Development environment
- MySQL Workbench 8.0 — Initial attempt (superseded by DuckDB)